

colomboType #001

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on 7th April 2018, Colombo

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Hi, I am

Sumanthri

I love to research on typography

Why?

i wanted to design a font - Sinhala font

more... to the Why?

i am a lecturer ---------- what about Sinhala script?
it was time to do my higher studies ---------- it was time!

PhD

i was searching for data - scientifically speaking - Researching

every where I go

ended at a dead end - a white wall!

PhD

i was searching for data - scientifically speaking - Researching

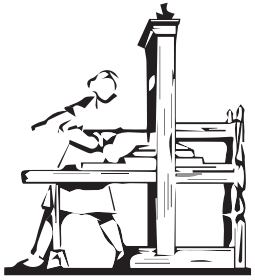
ONLY WAY OUT WAS TO dig in

found letters - but some were already found - up to the 15th Century



the letters after the 16th century

this is me!



The start of the typographic era in Sinhala script

“So, if someone wanted to design a Sinhala font.... wat aspects should i address”

the root letters for Sinhala
the distinguished Characteristics of each letter
the visual variations of Sinhala fonts
the historical aspect
the technological changes, the designers - the meta data

This is what I did!



It was a start of a new Journey

Root letters for Sinhala: Minimal letters with maximum properties

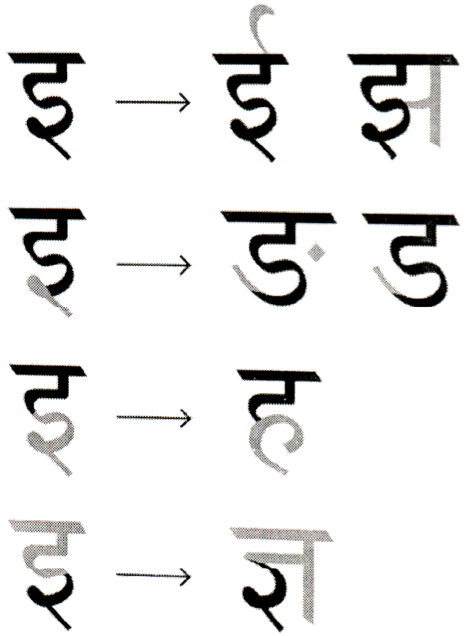
Dr. Sumanthri Samarawickrama
Dept. Integrated Design, University of Moratuwa, **Sri Lanka.**

Concept of Root letters

‘root letters’ communicate an
Indian philosophic concept of ‘*bijakshara*’

(*bija* meaning seed and *akshara* meaning letter);
and hypothesizes a set of primordial letters
which contain the germ of an entire font
(G. Dalvi 2010: 238).

Derivation of letters



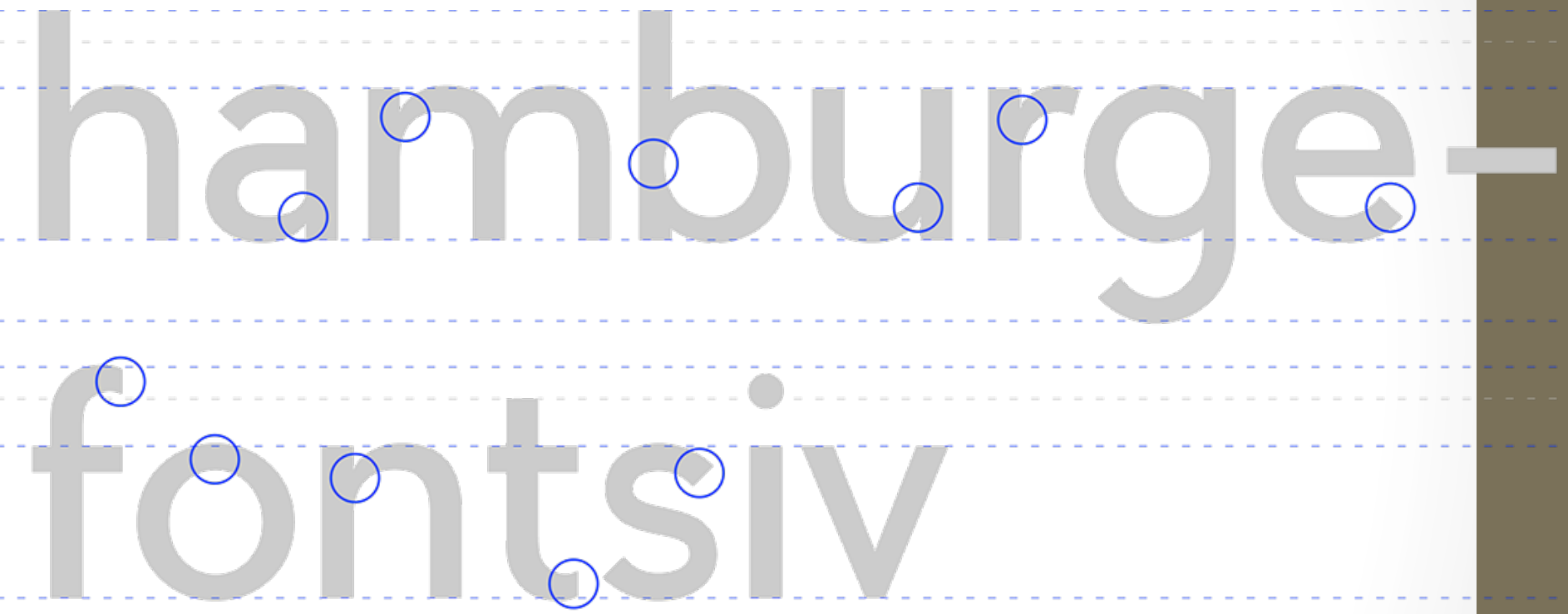
h - l n m u

Devanagari Script

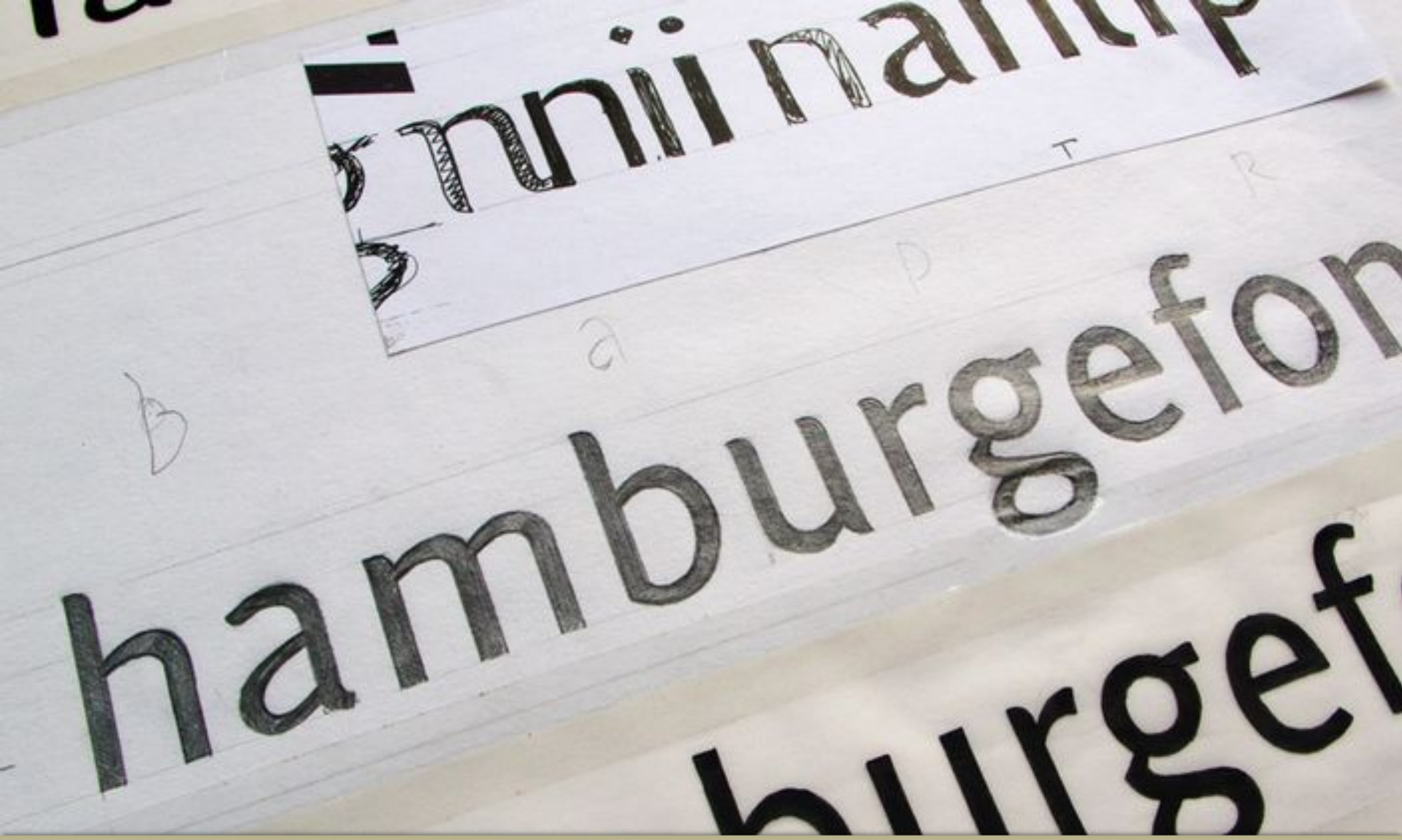
Latin Script

The composition of other letters derived from one single letter derivation

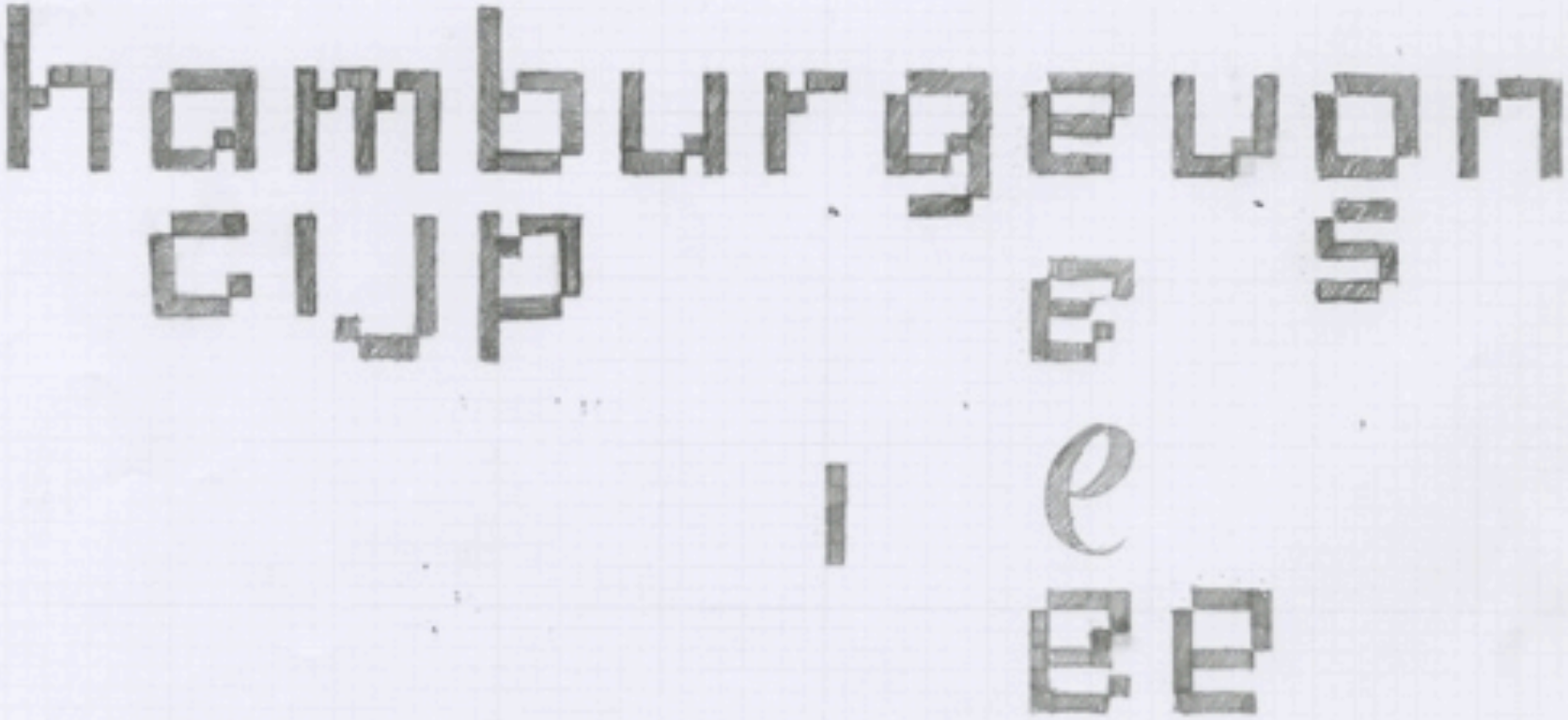
Dalvi's calculation method is a pioneer experiment and the only proposal applied to Indic script; therefore the author testes it on the Sinhala script



Hamburgefontsiv



Hamburgefonts



Hamburgevon


ADHESION THIN
ADHESION ROMAN
ADHESION BOLD
ADHESION BLACK
ADHESION ROUND
ORIGINAL TYPEFACE | ALLONE

Adhesion

Dalvi's theory

'Minimal letters with maximum properties' used to generated root letters for Devanagari

Dalvi's model

Tests on semi experts, letter properties and a scoring mechanism to calculate a score for each letter.

objective

To identify the root letters for the Sinhala Script.

The sub-objectives :

- To understand how much Dalvi's calculation method is effective when applying for Sinhala.
- What are the factors associated with Dalvi's calculation method and its influence on the final result.

Sinhala Script

- The type design process had evolved within a limited number of **self-taught professionals**.
- The letter properties were **not identified.**





Variables

The letters? the visual features?

Variables according to Dalvi

- Arrived at observing 50 font samples.
- Typefaces were partially random and partially selective,
- The concluding variables were based on clusters that were interpreted by experts and non-experts and the selected variables: counters, turns, knots, grey value, vertical terminals, contrast, axis, broad/tall, tool and the inclinations.

Dalvi describes these properties as **composite properties** as the variables are conceptualized according to the findings (in an initial study) through a visual survey.

Scoring system					
Variables	Property 01	Pr. 02	Pr. 03	Pr. 04	Pr.05
Letters					
A					
B					
C					
D					
E					
F					

Sample

Variables: for Sinhala?

- The type design process had evolved within a limited number of **self-taught professionals**.
- The letter properties were **not identified.**

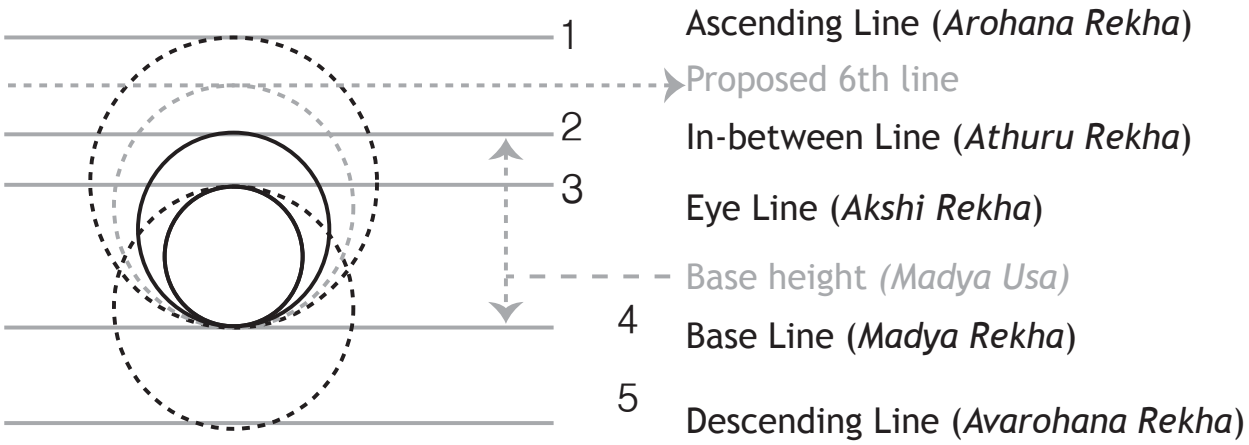


Variables based: proposed grid

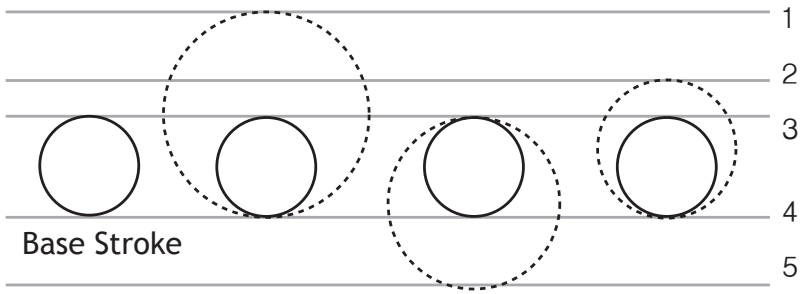
Identified by the author and published

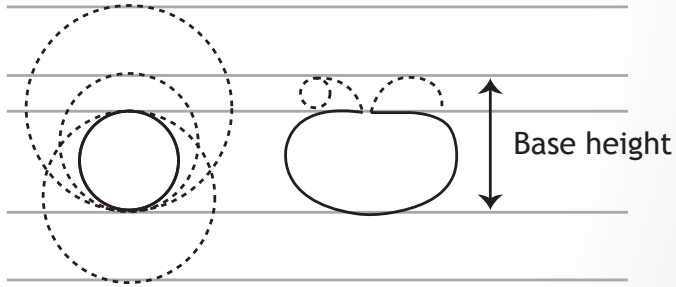
(Face Forward Conference, Dublin- Dec 2015).

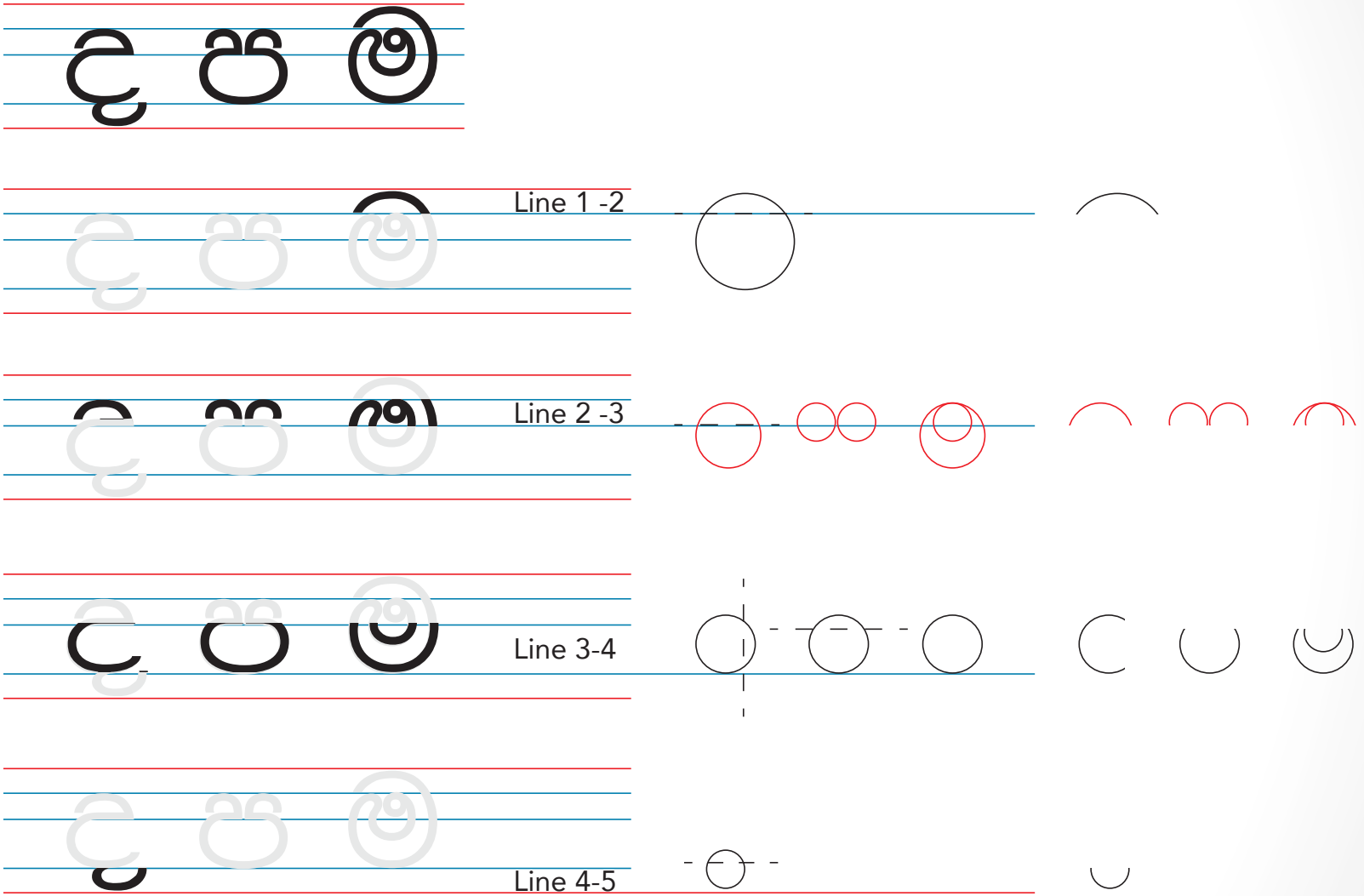
- Visual survey- using an action research spiral. Which listed out distinguished features (DF) on the five guidelines.
- One typeface -The typeface introduced in the NIE publication (2005)



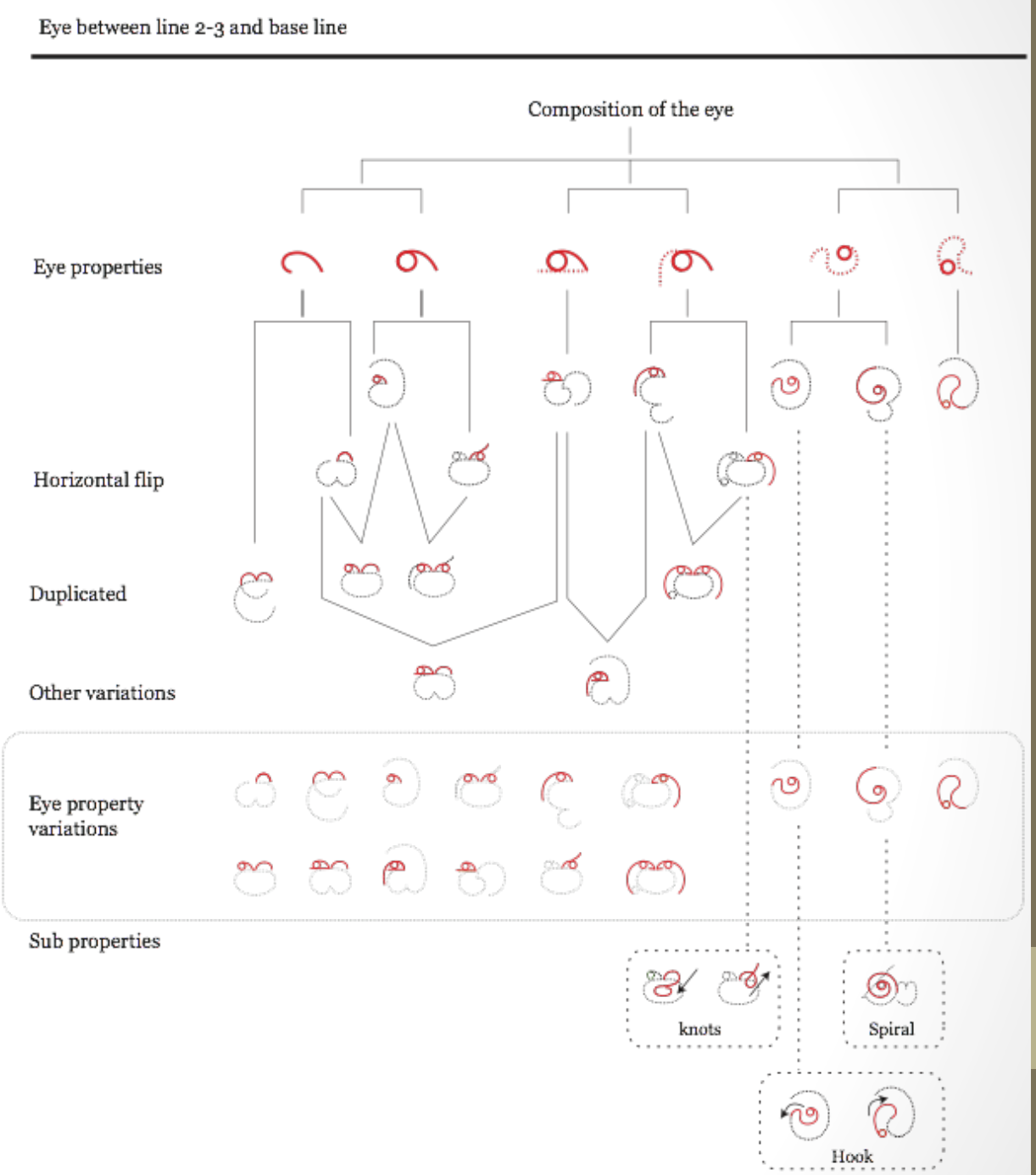
Ascender and Descender Stroke in relation to the Base Stroke

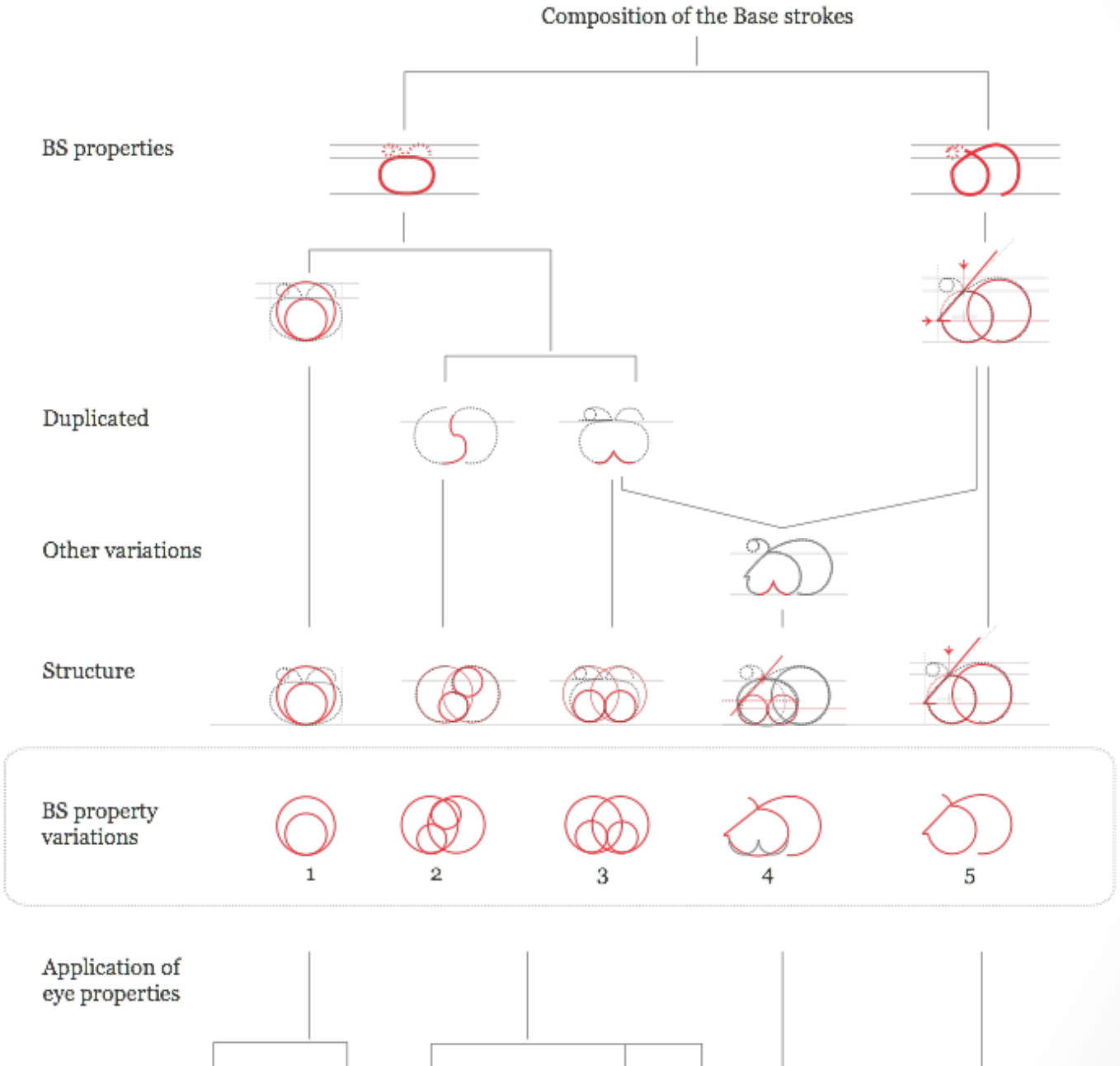




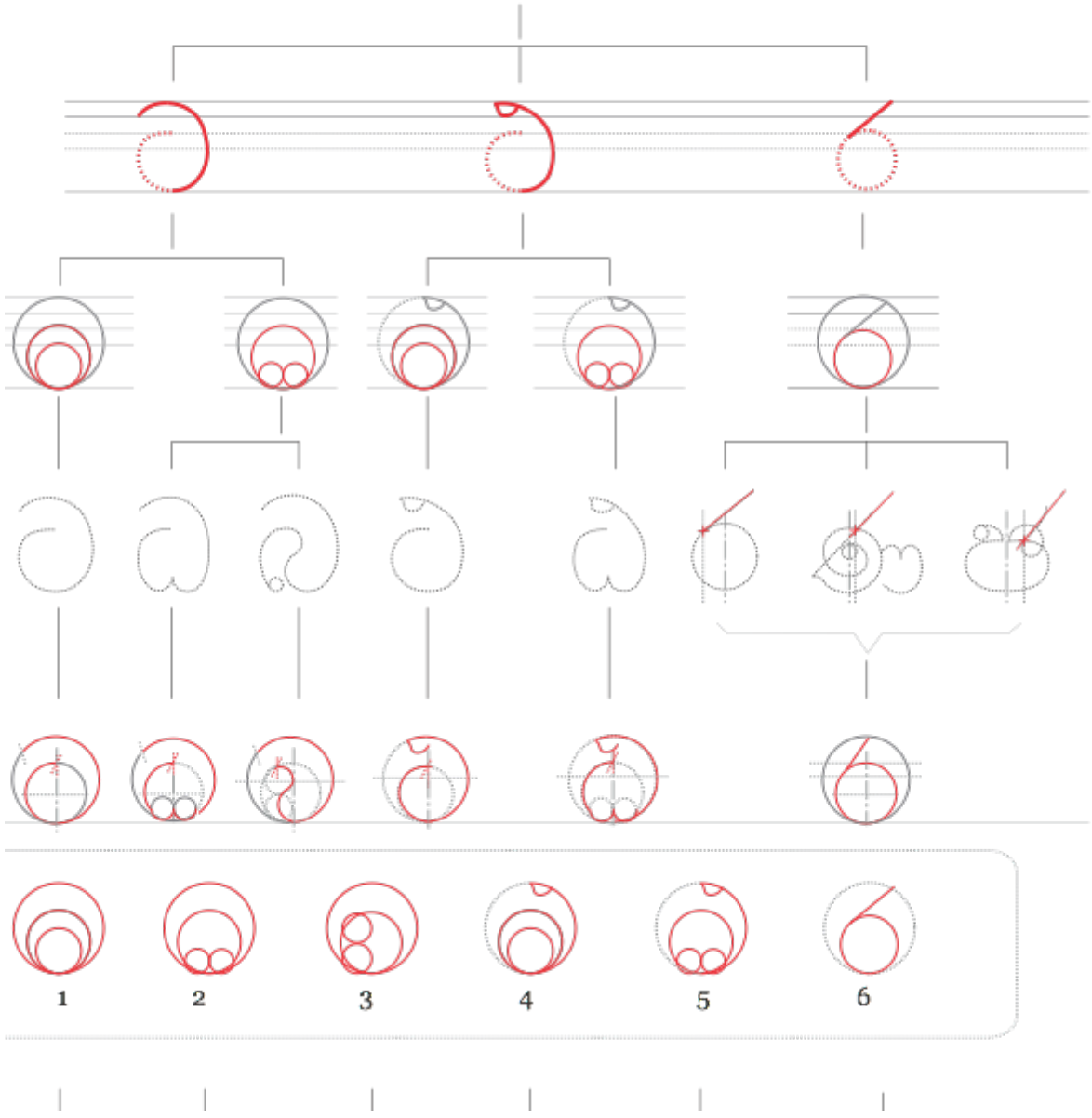

Line 1 -2
Line 2 -3
Line 3-4
Line 4-5

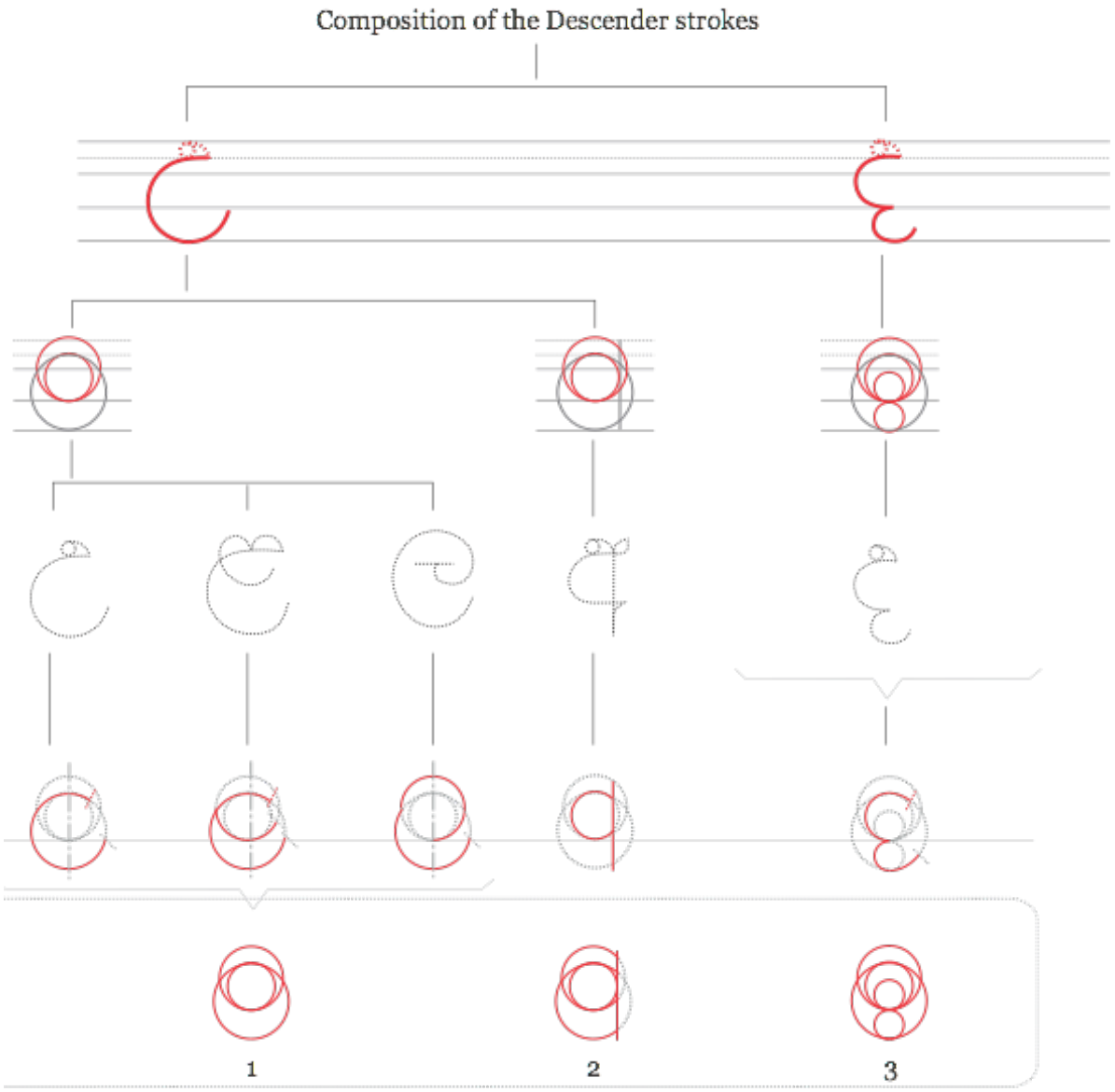
THE EYE


Eye between line 2-3 and base line
Composition of the eye
Eye properties
Horizontal flip
Duplicated
Other variations
Eye property variations
Sub properties
knots
Spiral
Hook


Composition of the Base strokes
BS properties
Duplicated
Other variations
Structure
BS property variations
1
2
3
4
5
Application of eye properties

Composition of the Ascender strokes



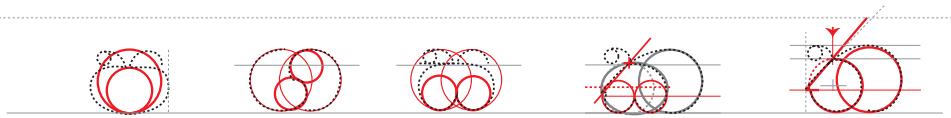

Composition of the Descender strokes
1
2
3

Eye property variations



1 2 3 4 5 6

Structure

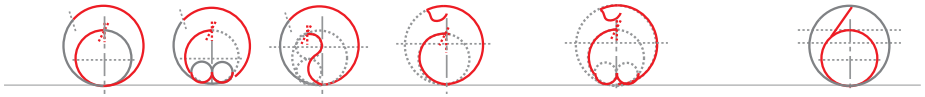


BS property variations



1 2 3 4 5

Structure

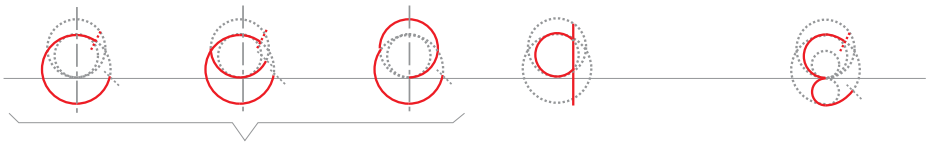


Ascender property variations



1 2 3 4 5 6

Structure



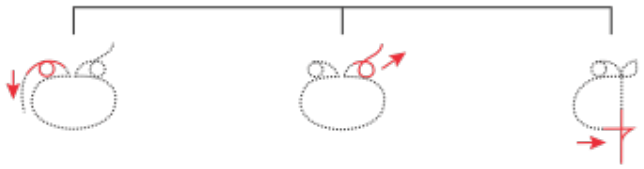
Descender property variations



1 2 3

Visual properties of the DF : Knot

Knot

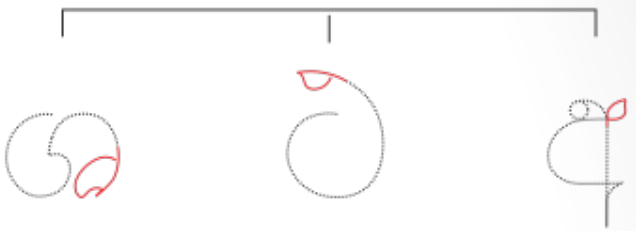


Variables composed with the DF : Knot



Visual properties of the DF : Loop

Loop



Variables composed with the DF : Loop



Visual properties of the DF : Hook/ Link

Hook/ Link



Variables composed with the DF : Hook/ Link



Visual properties of the DF : Stem

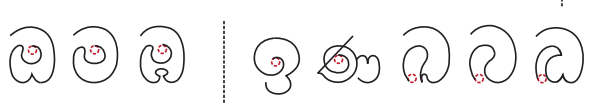
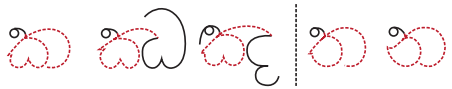
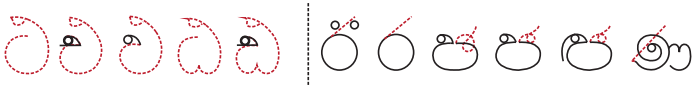
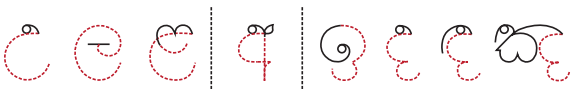
Stem



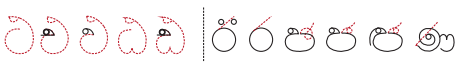
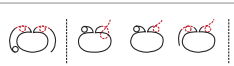
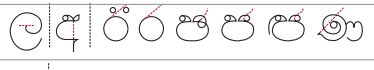
Variables composed with the DF : Stem



List of distinguished features/ properties and the varibales

DF	Properties		Variables
1. Eye	Eye 01		
	Eye 02		
	Eye 03,04		
	Eye 05,06	06	
2. Base Stroke	BS 01, 02, 03		
	BS 04, 05	11	
3. Ascender Stroke	AS 01, 02, 03, 04		
	AS 05 As 06	16	
4. Descender Stroke	DS 01, 02, 03,	19	
5. Knot	01, 02, 03,	22	

List of distinguished features/ properties and the varibales

DF	Properties		Variables
1. Eye	Eye 01		
	Eye 02		
	Eye 03,04		
	Eye 05,06	06	
2. Base Stroke	BS 01, 02, 03		
	BS 04, 05	11	
3. Ascender Stroke	AS 01, 02, 03, 04		
	AS 05 As 06	16	
4. Descender Stroke	DS 01, 02, 03,	19	
5. Knot	01, 02, 03,	22	
6. Hook/ Link	01, 02,	24	
7. Nose/ Point	01, 02,	26	
8. Hump	01, 02,	28	
9. Stem	01, 02, 03	31	
10. Spiral	01, 02,	33	
11. Shoulder	01, 02, 03	36	
12. Loop	01, 02, 03	39	

No.	Variables	DF	eye						Base Stroke					Ascender	
		Properties	eye 1	eye 2	eye 3	eye 4	eye 5	eye 6	BS 1	BS 2	BS 3	BS 4	BS 5	As 1	As 2
1	අ			1											
2	ඉ							1							
3	ර														
4	ට			1											
5	සා		1	1							1				
6	ෆ					1			1						
7	ච				1										
8	ම						1								
9	ක			1								1			
10	ඩ							1							
11	ග									1					
12	ඝ		1		1						1				
13	ඩ							1							
14	ඟ									1					
15	ච				1									1	
16	ඡ		1	1											
17	ජ			1											
18	ඣ			1								1			

The composition of other letters derived from one single letter derivation

The unique letters (UL)

The total score of the properties were summoned up and placed in ascending order.

The properties that have the lowest score were isolated, as those atomic properties would only be relevant to the letters that had the lowest score.

1.0	Unique letters in Sinhala
අ, ඖ, ඕ, ල, ඹ, ඉ, ණ, ඟ, ළ, න	

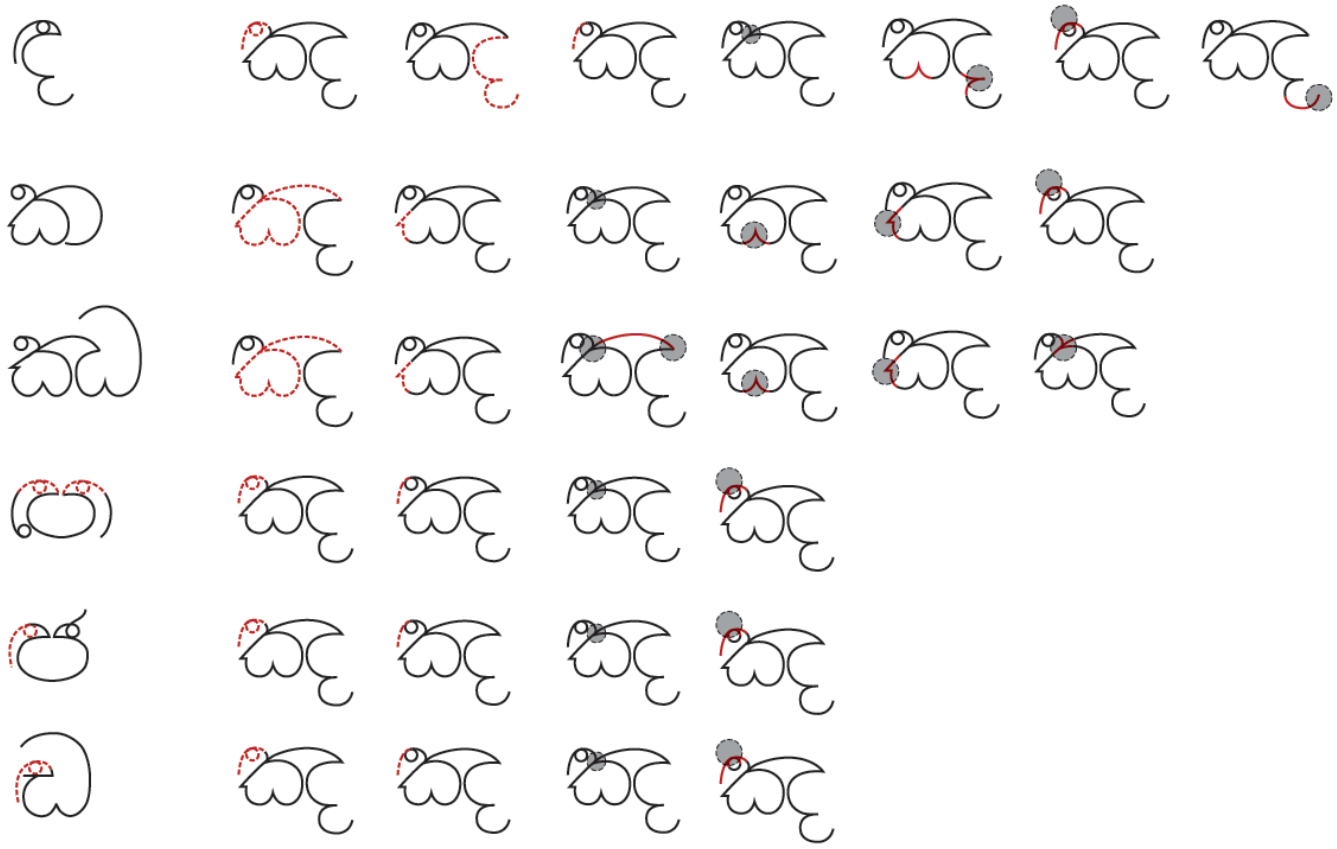
Root letters for Sinhala: evaluation	Unique letters	Max letters - range	Max properties
RL 01 – initial calculation	3	05 -15	NA
RL 01 *nya* (ඤ), *nja* (ඥ), *iru* (ඍ), *nda* (ඬ), *kha* (ඛ), *tha* (ත), *o* (ඔ), *ndha* (ඣ), *da* (ද), *bha* (භ)	0	20 -30	40/49

Letters derived from the Root letter : 1

d.



ද ඛ ඞ ජ ඪ ඉ ණ බ ළ ඍ ඝ ඡ ය ඓ



Letters derived from the Root letter : 1

d.



ද කඩ ජඇඳ ඉදන ණබළ සෑඩසයඟ



Root letter : 1 selection of first three letters



ද කා ඩඩ

Root letter : 1

a.



Eye 04, Nose/Point 01, Base stroke 04, Shoulder 03, Decsending stroke 03,

---

Eye joint 04, Right angular J, Intersection 02, Curve to C Joint 02, Curve andgular J 01, Terminal 08, Curve to C Joint 04, Intersection 01,

---

b.

Eye 04



Base Stroke



Descender Stroke



Nose/ Point



Shoulder



Root letter : 1

a.



Eye 04, Nose/Point 01, Base stroke 04, Shoulder 03, Decsending stroke 03,

Eye joint 04, Right angular J, Intersection 02, Curve to C Joint 02, Curve andgular J 01, Terminal 08, Curve to C Joint 04, Intersection 01,

b.

Eye 04 

Base Stroke 

Descender Stroke 

Nose/ Point 

Shoulder 

Eye Joint 04 

Curve to Curve Jt 02 

Curve to Curve Jt 04 

Right angular Jt 

Curve angular Jt 01 

Intersection 01 

Intersection 02 

Terminal 08 

c. Letters derived from the Root letter : 1



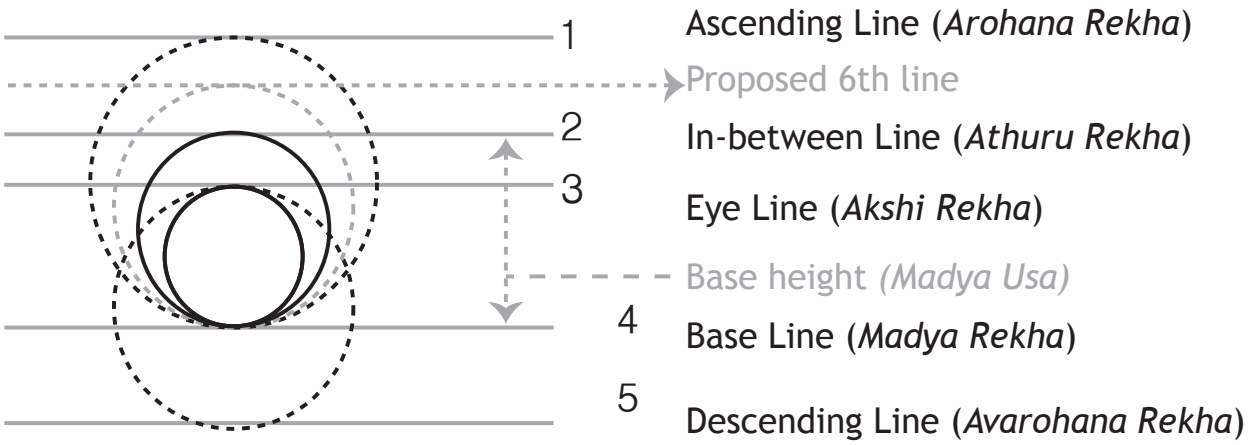
ඳ ඛ ඬ ඡ ඣ ඉ ණ ළ ඍ ඤ ඥ ඟ

Sample card of root letter *kna:*

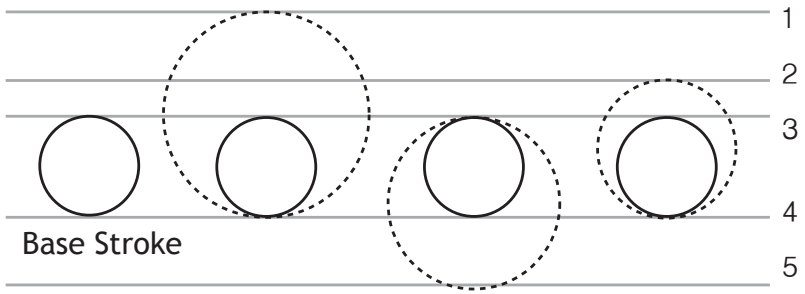
its visual properties and common letters

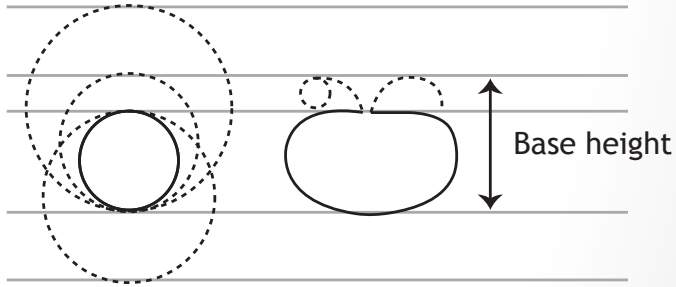
Testing the Distinct features

With the proposed grid method



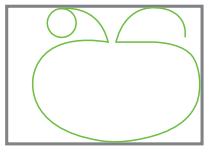
Ascender and Descender Stroke in relation to the Base Stroke

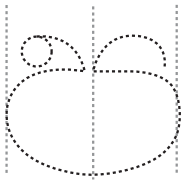


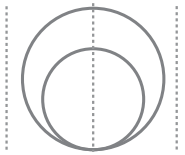


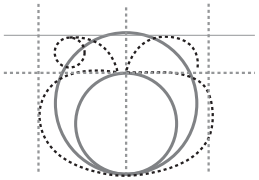
Composition of the Base stroke 01

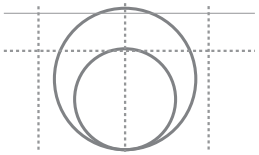
ga ration

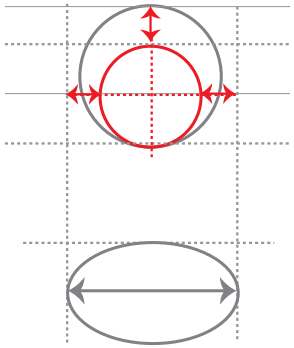


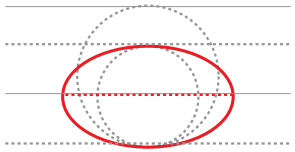




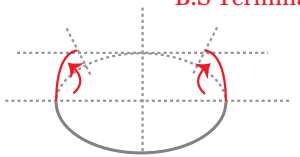


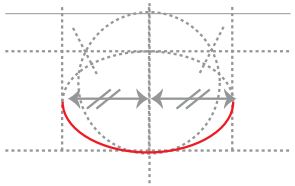




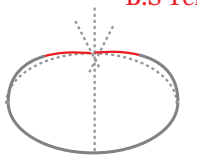


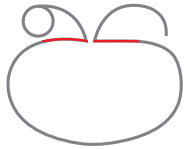
B.S Terminal 01



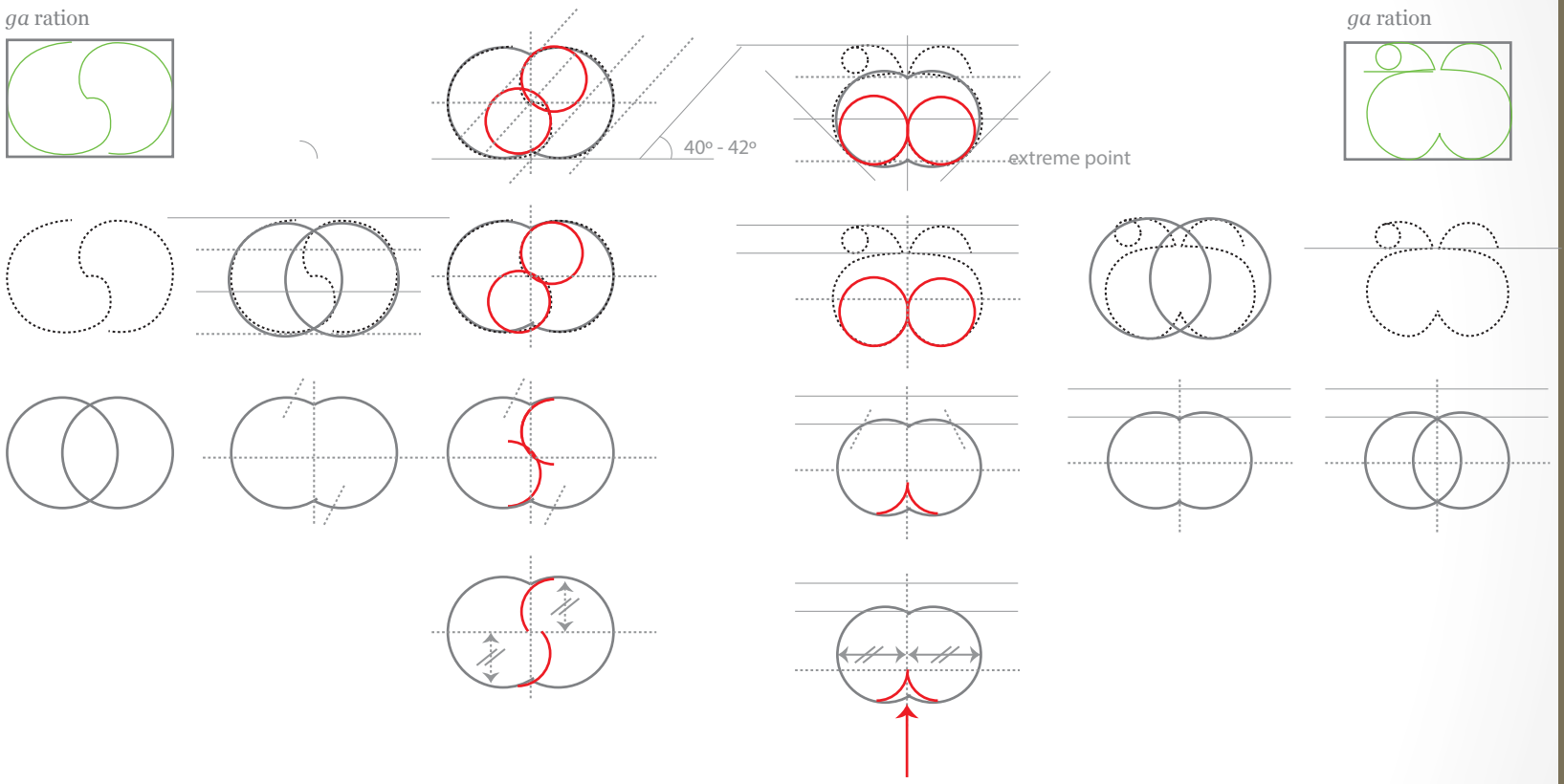


B.S Terminal 02

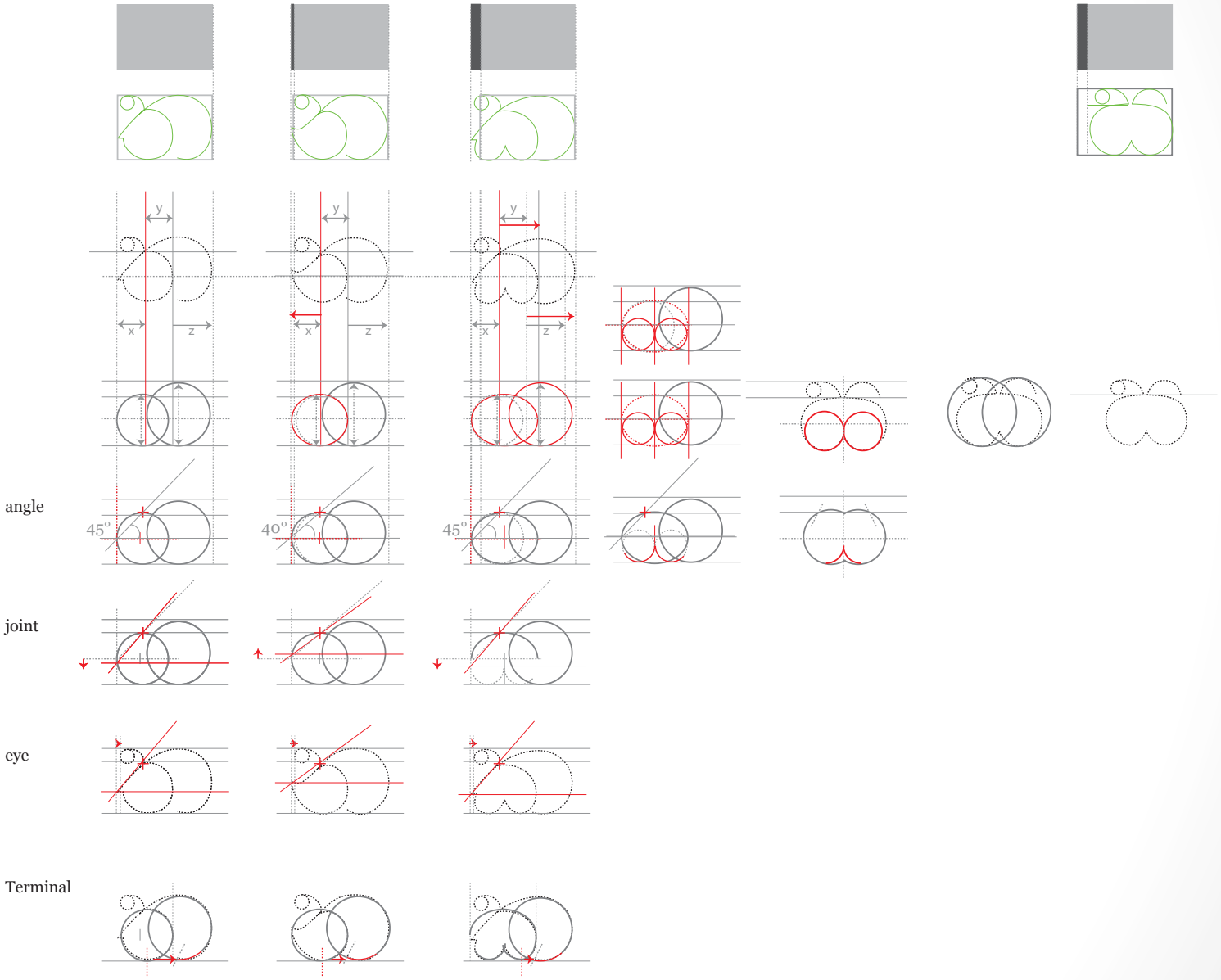




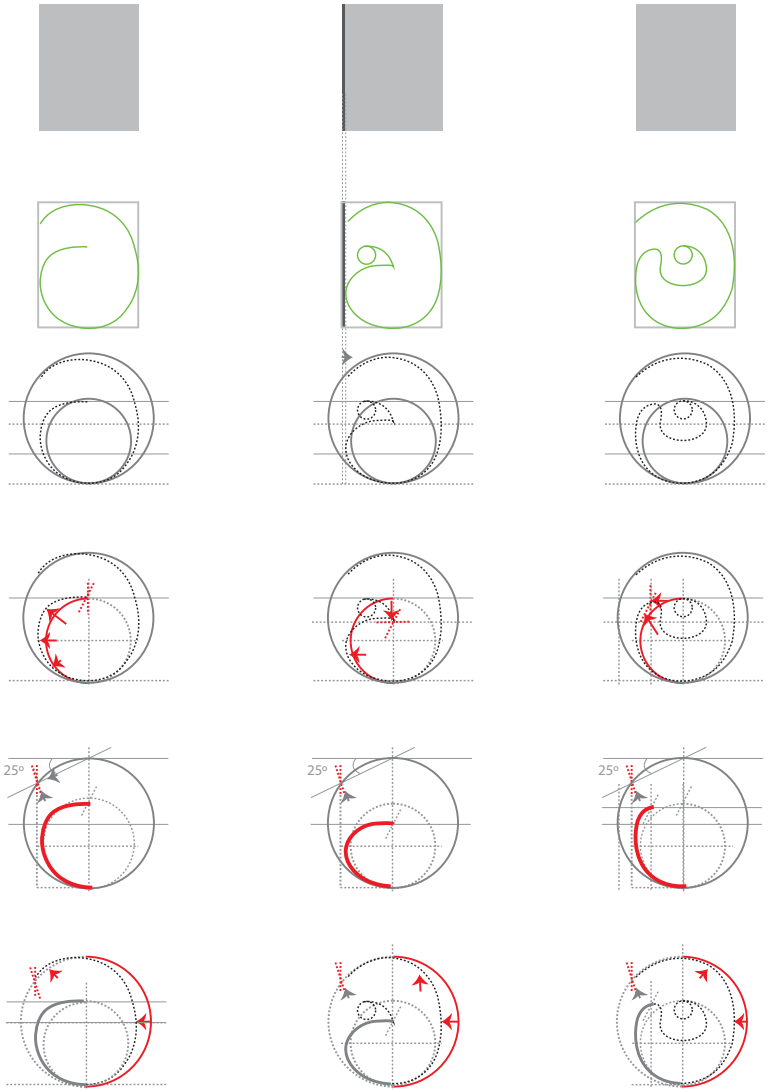
composition of the Base stroke premitive 02 an 03



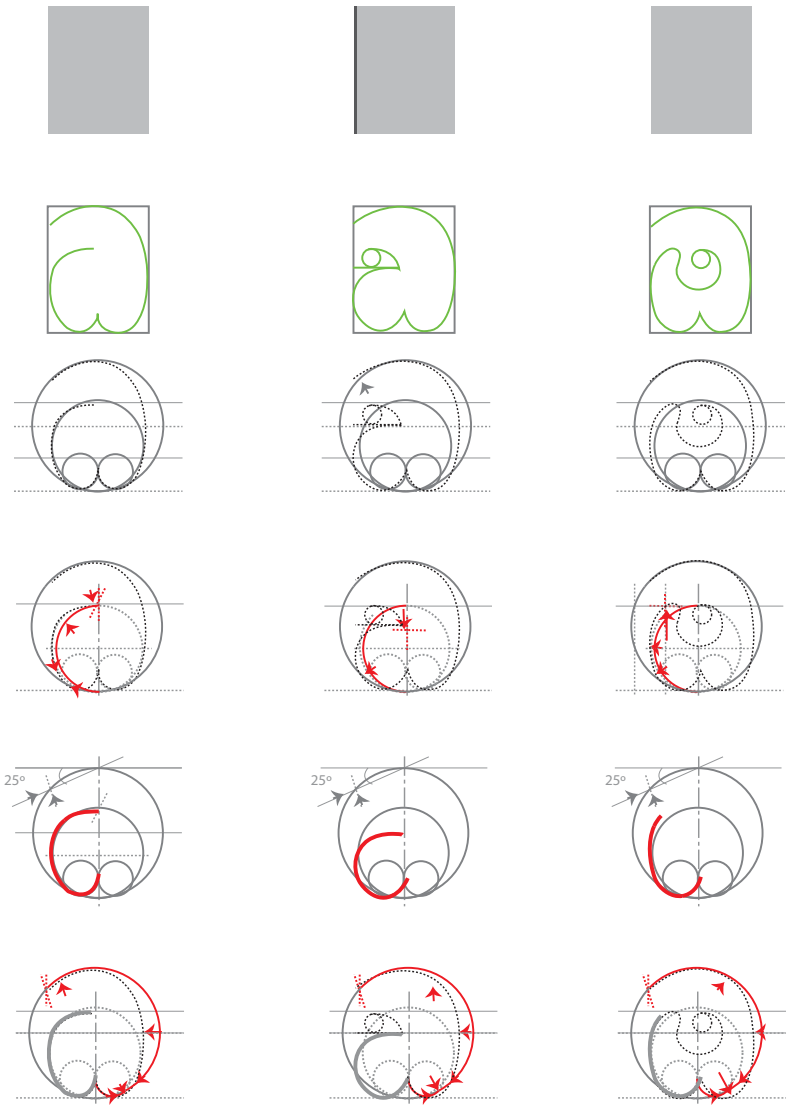
Composition of the Base stroke premitive 03 and 05



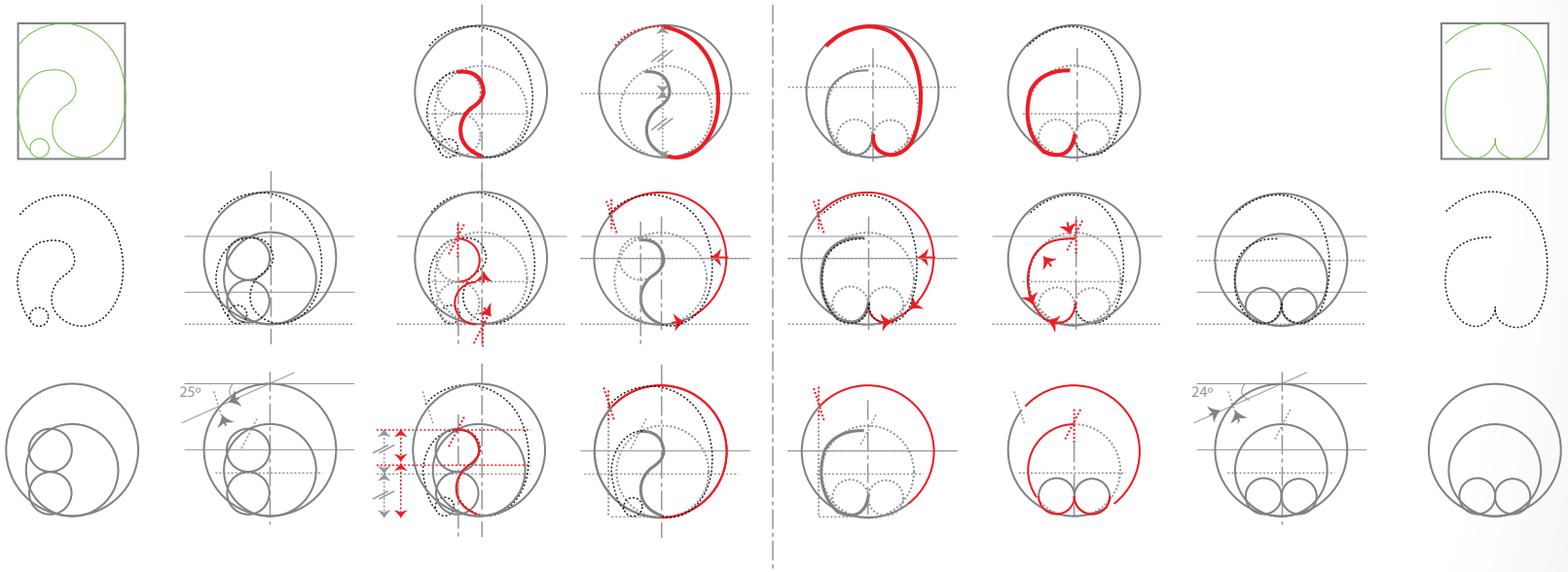
Composition of the ascender stroke premitive 01



Composition of the ascender stroke premitive 02



Composition of the ascender stroke premitive 02 and 03



Root letters for Sinhala: evaluation	Unique letters	Max letters - range	Max properties
RL 02 *pa* (ප), *g a*(ග), *ka* (ක), *ma* (ම), *nda* (ඬ), *ba* (ඹ), *ta* (ඨ), *ra* (ර), *u* (උ), *da* (ද).	0	20 -30	41/49

	Unique letters	Max letters - range	Max properties
RL 01 –initial calculation	3	05 -15	NA
RL 01	0	20 -30	40/49
RL 02	0	20 -30	41/49

Root letters for Sinhala: evaluation	Unique letters	Max letters - range	Max properties
RL 01 – initial calculation	3	05 -15	NA
RL 01 *nya* (ඤ), *nja* (ඥ), *iru* (ඍ), *nda* (ඬ), *kha* (ඛ), *tha* (ත), *o* (ඔ), *ndha* (ඣ), *da* (ද), *bha* (භ)	0	20 -30	40/49
RL 02 *pa* (ප), *g a*(ග), *ka* (ක), *ma* (ම), *nda* (ඬ), *ba* (ඛ), *ta* (ඨ), *ra* (ර), *u* (උ), *da* (ද).	0	20 -30	41/49

Findings and observations

- Root letters
- Dalvi's calculation method
- The proposed grid method
- Unique letters for Sinhala
- Appropriate set of root letters for Sinhala, type design process